

IGAS.

gas analytical
equipment



Automation cabinet «IGAS» passport

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Appendix 1. IGAS automation cabinet statement of work

Appendix 2. Power consumption calculations.

Appendix 3. Communication protocols

1. General Description

The Cabinet is used to continuously monitor remote sensor transmitters in order to warn of potentially dangerous or harmful conditions. Cabinet based on a modular PLC IGAS LCP2xxx and modules IGAS LXXxxx. Other differences are explained in this document where necessary. Cabinet can handle a wide range of gas detectors via current loop or RS485 through analogue inputs

It can control various outputs such as audible and visible signals, solenoid valves, venting systems etc. All events transfer via Modbus or IEC60870-5 -104 to control room or SCADA.

Every cabinet has a service USB connector for configuring and diagnostic.

The controller monitors the current signals from the remote sensor transmitters on input channels and provides warnings when alarm or fault conditions occur. Each of the input channels can provide alarms for up to two gas levels and one

fault condition. The gas-level alarms are configured to provide an indication of potentially dangerous or harmful gas levels at the transmitter. The fault operate when the transmitter sends the relevant fault signal.

The combination of internal control modules (input modules, relay modules and output modules) and the controller configuration settings can be altered to meet the gas detection and warning requirements.

Additionally the warning devices used with the controller can be selected and configured as required by the user. The cabinet is configured using a Microsoft Windows® based software program. The touchscreen also has a built-in configuration system.

The cabinet is constructed from high-impact steel and have fully-sealed, easy opening access. It is supplied with a wall mounting or can be directly mounted to any solid vertical surface. Cable entry is via rubber entry on the lower side.

2. Technical discription

The Cabinet is designed to operate on a single phase, 50 to 60 Hz, 100 ... 240 VAC supply with a maximum power consumption of less than 100 W (depends upon configuration).

Cabinet can contain an optional backup battery to guard against short-term power disruption.

The cabinet are constructed from powder coated steel with a quick release front access door panel. The sytem door panel holds a touch sensitive colour LCD and 4 status LEDs. Inside the cabinet is a din rail installed Switched-Mode Power Supply providing a nominal DC 24 V output, an optional Lithium-ion backup battery, a Main CPU Module, two Relay Output Modules, protection circuit breakers, automatic phase selector and input/output terminals.

Enclosure contain a common Earth (ground) rail that must be connected to Protective Earth (Ground) .

3. Condition of Use

Parameter	Value
Ambient Operating Temperature	
standard cabinet	+1 °C to +55 °C
with heater option	-60 °C to +50 °C
Operating Humidity (non-condensing)	5 % to 95 %RH
Storage	
Without batteries	-25 °C to +60 °C; 5 % to 95 %RH, non-condensing
With batteries	
-20 °C to +25 °C	1 Year
-20 °C to +45 °C	3 Month
-20 °C to +60 °C	1 Month
IP Rating	IP65

4. Dimensions

See Appendix 1.

5. Power requirements

Parameter	Value
Input Voltage	~ 100...240 VAC
AC Frequency Range	50 – 60 Hz ± 6%
DC Voltage Output	24 VDC
Rated Power Output	see Appendix 2.

6. Communication protocol.

See appendix 3.

7. Items checklist

Name	Number	Quantity
Automation cabinet «IGAS»	КДЦА.422412.001	1
Cabinet Passport «IGAS»	КДЦА.422412.001 ПС	1
Instruction for use Automation cabinet «IGAS»	КДЦА.422412.001 ПС	1

8. Warranty

The manufacturer guarantees the compliance of the gas analyzer with the requirements of the IGAS technical specifications TY 3431-002016777305-13.

Warranty void if gas analyzer operate, stored or transported out of specification limits

Warranty period 60 months

Manufacturer IGAS Engineering

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10. Verification certificate

Automation cabinet «IGAS» serial number _____

verified refer to TY 3431-002016777305-13 and fully ready for operation.

Date of manufacture _____

IGAS Quality control manager _____
signature / stamp

11. Sales data

Date of sale _____

Sales manager _____
signature / stamp

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